

Quantum Mechanics-I

The University of Iowa
The College of Liberal Arts and Sciences
Fall, 2026

Description of Course

This course is an advanced Quantum Mechanics course for graduate students. Topics to be covered include Chapters 1 – 4 of Sakurai & Napolitano and applications of quantum mechanics. Please see the course calendar for more details. This course is a requirement for PHYS:5742.

Learning Objectives

Students will gain a deeper understanding of Dirac notation, the Schrödinger equation and its elementary solutions, and solve advanced problems in dynamical quantum systems. Students will also gain greater familiarity with spin and angular momentum, tensor operators, symmetries, and the density matrix formalism.

Outline of Course Topics

- **Module 1:** Introduction and Formalism
Dirac notation and operators, measurements, position and momentum, and wavefunctions.
- **Module 2:** Time evolution
Time evolution operator, Schrödinger's equation and solutions, harmonic oscillator, propagators, and path integrals.
- **Module 3:** Angular Momentum
Rotations, spin and angular momentum operators, Hydrogen atom, addition of angular momentum, tensors and Wigner-Eckart theorem.
- **Module 4:** Ensembles and Density Matrix
Density operators, product and mixed states, and open systems.
- **Module 5:** Symmetries & Overflow
Symmetries in central potentials, parity and selection rules, and time reversal.

Required Textbooks, Readings, and Other Materials

The required textbook(s) for this course are:

Modern Quantum Mechanics (edition: 3rd) Required
ISBN: 9781108473224

Author: Sakurai
 Publisher: CAMBRIDGE UNIV PRESS. ©2020
 Approximate Price: \$54.75

Types of Assessments

Weekly homework assignments, two midterm exams, and one final exam.

Course meeting time and place

9:30 am -- 10:20 am Mondays, Wednesdays, Fridays, VAN 161

Department of Physics & Astronomy

<https://physics.uiowa.edu/>

Instructor: Prof. Matheus Hostert
Phone: 319-335-2943
Email: matheus-hostert@uiowa.edu
Office location: VAN 502

Student drop-in hours

11:00 am -- 12:00 am Tuesday
 3:00 pm -- 4:00 pm Wednesday
 2:30 pm -- 3:30 pm Thursday
 Or by appointment

Departmental Executive Officer (DEO): Gregory G. Howes (gregory-howes@uiowa.edu)

Student Complaints

Students with a complaint about a grade or a related matter should first discuss the situation with the instructor and/or the course supervisor (if applicable), and finally with the DEO (Chair) of the department, school or program offering the course. Sometimes students will be referred to the department or program's Director of Undergraduate Studies (DUS) or Director of Graduate Studies (DGS).

Undergraduate students should contact [CLAS Undergraduate Programs](#) for support when the matter is not resolved at the previous level. Graduate students should contact the [CLAS Graduate Affairs Manager](#) when additional support is needed.

Course's College (Administrative Home)

For undergraduate courses

The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the add and drop deadlines, academic misconduct policies, and other undergraduate policies and procedures. Other UI colleges may have different policies.

For graduate courses

The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the policies and procedures for its courses. Graduate students, however, must adhere to the [academic deadlines set by the Graduate College](#).

Drop Deadline for this Course

You may drop an individual course before the drop deadline; after this deadline you will need collegiate approval. You can look up the drop deadline for this course [here](#). When you drop a course, a “W” will appear on your transcript. The mark of “W” is a neutral mark that does not affect your GPA. To discuss how dropping (or staying in) a course might affect your academic goals, please contact your Academic Advisor. Directions for adding or dropping a course and other registration changes can be found on the [Registrar’s website](#). Undergraduate students can find policies on dropping CLAS courses [here](#). Graduate students should adhere to the [academic deadlines](#) and policies set by the Graduate College.

UI Email

Students are responsible for all official correspondences sent to their UI email address (uiowa.edu) and must use this address for any communication with instructors or staff in the UI community. For the privacy and the protection of student records, UI faculty and staff can only correspond with UI email addresses.

Course ICON site

To access the course site, log into [Iowa Courses Online \(ICON\)](#) using your Hawk ID and password.

Grading System

Final grades will be awarded based on the following ranges:

A		B		C	F
A+	97-100	B+	87-83	C+ 70-60	<60
A	97-92	B	83-80	C	
A-	92-87	B-	80-70	C-	

Grading System

Possible final grades are A+ to F.

B represents a minimal satisfactory learning, **A** an outstanding performance, and **C** and below a substandard performance indicating that the completion of a PhD degree may be difficult. Plus/minus grades may be given. No curve involved. **A+** grades are given only in

extraordinary situations.

Homework assignments, solutions, and notes will be posted on ICON.

Course Grades

Homework 30%, Midterm-1 20%, Midterm-2 20%, Final Exam 30%.

Homework assignments will be set every week with exceptions to be announced on ICON. Each homework contributes equally to the final homework grade. Homework will be self-graded, and the grading will be reviewed by the instructor.

Homework assignments will be set on Fridays and will be due on the next **Friday by 11:59 pm**. Please check announcements on ICON for more details.

Late submissions will be penalized by 10 percentage points within the first 24 hours, and 20 percentage points within the second 24 hours. No assignments will be accepted after 48 hours, including weekends.

Attendance and Accommodations

Regular attendance is encouraged but not enforced. While attendance is not required, you will be responsible for all the material covered in the lectures.

Questions during class are encouraged, no matter how trivial, however, the instructor may recommend some discussions to be continued during office hours. Please bring your own writing material to solve exercises during class.

[Absences from Class](#)

[Absences for Religious Holy Days](#)

[Absences for Military Service Obligations](#)

[Accommodations for Students with Disabilities](#)

Students should feel free to get in touch via email or during office hours to discuss any accommodations necessary, as outlined above.

Other Expectations of Student Performance

[Free Speech and Expression](#)

[Non-discrimination Statement](#)

[Classroom Expectations](#)

[Class Recordings](#)

Academic Honesty and Misconduct

All students in CLAS courses are expected to abide by the [college's standards of academic honesty](#). Undergraduate academic misconduct must be reported by instructors to CLAS

according to [these procedures](#). Graduate academic misconduct must be reported to the Graduate College according to Section F of the [Graduate College Manual](#).

Collaborative work is encouraged, especially for homework assignments. Plagiarism is not acceptable. Students are responsible for ensuring their own academic honesty and encouraged to ask their instructor for clarification if needed.

Artificial Intelligence (AI) Policy

Usage of AI tools such as ChatGPT, Copilot, or Claude, is allowed and encouraged for coding, idea generation, brainstorming, and latex transcription, but it should not be used to obtain full solutions to your homework. Note that hallucinations are common and can look convincing. AI tools should not be used for generating solutions or code you do not understand.

If AI is used in a homework or assignment, include a brief “AI usage” note explaining what you used it for. When appropriate, include reasoning explaining why you believe it is correct.

Date and Time of the Final Exam

Both midterms and the final exam will have a duration of 60 minutes.

1st Midterm: Wednesday, Oct 14th, 2026

2nd Midterm: Wednesday, Nov18th, 2026

Final exam: TBD

The [final examination date and time](#) will be announced by the Registrar generally by the fifth week of classes, and it will be announced on the course ICON site once it is known. **Do not plan your end of the semester travel plans until the final exam schedule is made public. It is your responsibility to know the date, time, and place of the final exam.** The [Registrar’s Office website](#) details which types of schedule conflicts qualify for make-up exams and the processes students should follow to request make-up exams. Not all conflicts qualify for make-up exam times, and requests must be made by the deadlines listed.

Calendar of Course Assignments and Exams

Please note this calendar is **a tentative plan and subject to change**. Ranges are inclusive.

Topics Covered	Dates (inclusive brackets)
Module 1: Introduction and Formalism	3 weeks
Dirac notation and operators, measurements, position and momentum, and wavefunctions. Labor day Sep 7 th (no class)	Weeks 1 – 3 [Aug 24 th – Sep 11 th]
Module 2: Time evolution	5 weeks
Time evolution operator, Schrödinger's equation and solutions, harmonic oscillator, propagators, and path integrals.	Weeks 4 – 8 [Sep 14 th – Oct 16 th] Midterm 1: Oct 14th (Wednesday)
Module 3: Angular Momentum	3 weeks
Rotations, spin and angular momentum operators, Hydrogen atom, addition of angular momentum, tensors and Wigner-Eckart theorem.	Weeks 9 – 11 [Oct 19 th – Nov 6 th]
Module 4: Ensembles and Density Matrix	2 weeks
Density operators, product and mixed states, and open systems. Fall Break (no classes in Week 14: Nov 22 nd – 29 th)	Weeks 12 – 13 [Nov 9 th – Nov 20 th] Midterm 2: Nov 18th (Wednesday)
Module 5: Symmetries & Overflow	
Symmetries in central potentials, parity and selection rules, and time reversal.	Weeks 15 – 16 [Nov 30 th – Dec 11 th]
Finals Week	Dec 14 th -18 th

Student Support Resources and Related PoliciesAcademic Support for this Course

I am available for help during drop-in hours, via email, or by appointment. Please feel free to reach out. You may also access academic support via Tutor Iowa at <https://tutor.uiowa.edu/> and writing support via The Writing Center at <https://writingcenter.uiowa.edu/>.

Some important resources are linked below.

[Mental Health](#)

[Basic Needs and Student Support](#)

[Sexual Harassment/Sexual Misconduct and Supportive Measures](#)

[Conflict Resolution](#)